

VENKATA SHANMUKHASAI K

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Software engineer building production AI platforms — agentic applications, personalization and decisioning, and distributed backends handling 10M+ records. AWS Certified Solutions Architect.

EDUCATION

State University of New York at Buffalo

Buffalo, USA

Master of Science in Computer Science, CGPA: 3.8/4.0

Aug 2024 - Feb 2026

- **Courses:** Data Models and Query Languages, Database Systems, Data Intensive Computing, Machine Learning, Operating Systems, Modern Network Concepts, Computer Security, Computer Architecture

PROFESSIONAL EXPERIENCE

National Center for Ontological Research — *applied AI research org*

Buffalo, USA

Software Engineer

March 2026 – Present

- Core developer of **Ontos Cosmos**, an applied-AI platform that turns any input — streaming data, audio, PDF, Excel, RDF/Turtle — into **governed, fully traceable** structured knowledge; generated **1,000+ knowledge graphs** across **9,400+ entities**, cutting expert modeling work from ~1 month to ~15 minutes.
- Single-handedly built core services across a **FastAPI** microservice backend, **React** frontend, **PostgreSQL**, and **GraphDB**, owning the platform end-to-end through **AWS** and **CI/CD** — including a schedulable **agentic pipeline** ingesting up to **15 GB** per run that combines **LLM, machine learning**, and semantic methods with **full provenance tracking** on every generated output.
- Built the platform's **automated evaluation and ranking engine** — every output is scored across 6 weighted dimensions and 30 checks, making them **explainable** and comparable — plus a registry of **400+ domain models** with hybrid keyword + **semantic vector search (Typesense)**, serving as the retrieval layer for **RAG**-grounded generation.

ProseWrite

Remote

AI Backend Engineer

May 2025 – March 2026

- Built the transcription service behind a product used by **5,000+ users** across author and academic communities — converting audio and video to text at **95%** accuracy with speaker attribution and **sub-2-minute** turnaround, plus **LLM** post-processing for transcript cleanup and summaries.
- Engineered the ingestion and processing pipeline to reliably handle **multiple 10 GB files**, with resilient retry handling and live progress streaming to the client.
- Extended the product from web-only to fully downloadable with a **Chrome Extension** (Manifest V3) for live meeting capture and a cross-platform **Electron** desktop app.

Bajaj Finserv Health — *digital health platform for payers, providers & patients*

Pune, IN

Software Engineer - II, Software Engineer - I

July 2022 – July 2024

- Built the **personalization rule engine** deciding what content and health messaging each user saw across homepage and wellness pages — rules evaluated against **MongoDB** with **Redis** caching, serving **300+ enterprise clients**.
- Made patient-record search roughly **60% faster** across **10M+ records** by separating reads from writes with **CQRS** and moving queries onto **Elasticsearch**, running on **Azure Kubernetes Service**.
- Owned 5+ **Node.js/TypeScript** services on a **Kafka** event-driven backbone — REST APIs, cron jobs, code reviews and monitoring — and led the **Docker** rollout across a 4-engineer team, shipping with zero downtime.

Bajaj Finserv Health

Pune, IN

Software Development Engineer Intern

January 2022 – June 2022

- Cut content-update dependency from 15 days to 2 days by delivering a CMS portal with **React** and a headless CMS.
- Contributed to **VoucherRx**, a voucher and rewards platform powering **100+ product integrations** — built redemption and rollback with **message-queue** architecture and **distributed locking**.

PROJECTS

- **Shop-Sphere: Microservice-Based Data Platform** 🛒 : Designed **event-driven data pipelines** using **Kafka** and **Redis** for real-time ingestion with **Elasticsearch**-powered product search; built scalable microservices in **Node.js** and **MongoDB** with automated test coverage, improving data reliability by 40%.
- **RapidMock Backend: A Dynamic API Mocking and Management Platform** 🛠️ : Built a layered **Express** backend (Repository pattern) with **MongoDB**, supporting mock endpoints, proxy forwarding, and multi-version response management; implemented schema validation, **JWT/bcrypt** auth, and project-level analytics with structured logging.

TECHNICAL SKILLS

- **Programming Languages:** Python, C++, C, Java, SQL, PL/SQL, Bash
- **Frameworks & Libraries:** Node.js, FastAPI, NestJS, Express.js, Spring Boot, gRPC, GraphQL, React, Angular
- **Databases & Infrastructure:** PostgreSQL, MongoDB, MySQL, Redis, Elasticsearch, Kafka, GraphDB, Typesense, AWS, Azure, Google Cloud, Docker, Kubernetes, Nginx, Linux, Git
- **Concepts & Practices:** Agentic AI, LLM APIs, RAG, Vector Search, Recommendation Systems, Personalization & Segmentation, Distributed Systems, Microservices, Event-Driven Architecture, CQRS, RESTful APIs, System Design, CI/CD, Concurrency, Load Balancing, Object Oriented Design, Design Patterns, Agile/Scrum

CERTIFICATIONS & PUBLICATIONS

- AWS Certified Solutions Architect – Associate (2025)
- Oracle Cloud Infrastructure GenAI Professional (1Z0-1127)
- *Lip Biometry on Handheld Devices*, IRJET (2021)
- *Machine Anomaly Detection*, Springer (2021)
- Databricks Certified Data Engineer Associate (2025)
- SnowPro Associate: Platform Certification
- *Phishing Detection using ML*, Smart Data Conf. (2021)